Inferring temporal shifts in a restored grassland from taxonomic, phylogenetic and functional diversity patterns

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The problem:

- *A robust restored ecosystem requires a functionally heterogeneous community of organisms with a broad range of traits that permits broad resource partitioning.*
- *Even though the effects of ecotypic variation or population sources of dominant species on community assembly have been widely studied, there is a lack of long-term studies on the crucial role of eco-evolutionary interactions in restorations across large-scale environmental gradients.*

The questions:

- *Would the home-site ecotype decrease community taxonomic diversity and have higher functional and phylogenetic clustering compared with exotic ecotypes?*
- *What do the effects of decomposing variability of each functional trait into interspecific and intraspecific trait variation show?*

Common garden in Carbondale, IL

Colors represent three sources of Andropogon gerardii and Sorghastrum nutans ecotypes sown with a prairie plant mix

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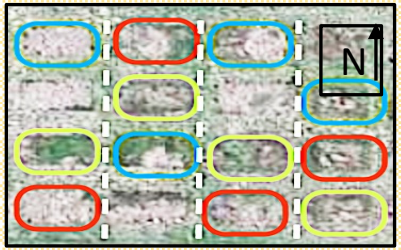

What we did:

Site: *A reciprocal common garden in Carbondale, IL.*

Study system: *Ecotypes of the two dominant species were seeded along with a prairie seed mix according to a randomized complete block design.*

Diversity: *Species composition was measured in 2012, 2014, 2018, and 2019 and taxonomic (richness), phylogenetic (PDsesmpd) and functional (FDsesmpd) diversity were calculated in each plot.*

Analyses: *Repeated LMMs for examining temporal shift; Variation in trait composition was decomposed into intraspecific trait variability, variability due to species turnover between communities and their covariation.*

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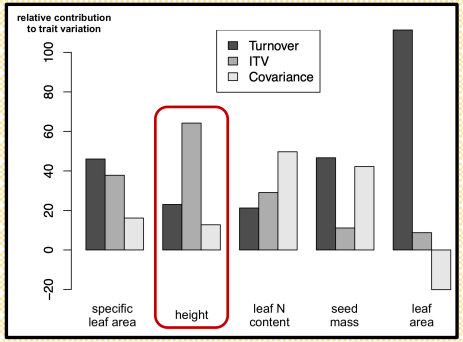
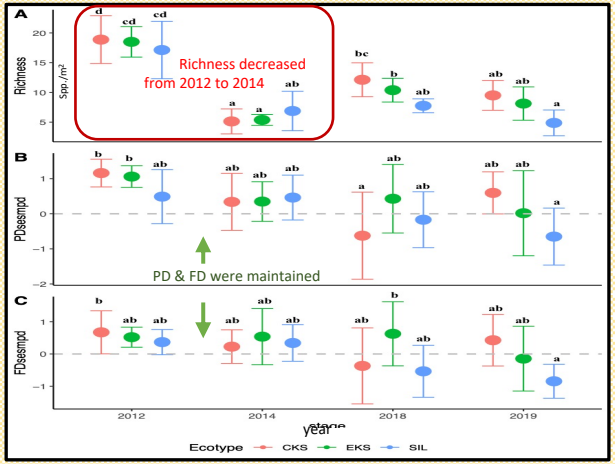
2nd: When intraspecific trait variation (ITV) was decomposed, the trait with the largest relative intraspecific variation was plant height.



What we found:

*1st: **Phylogenetic and functional diversity** were maintained at constant levels while **taxonomic diversity** declined during restoration.*



What it means:

- *Diversity patterns varied differently among metrics in restored grassland ---- phylogenetic and functional diversity were maintained at constant levels while taxonomic diversity declined during restoration.*
- *Intraspecific variation in crucial functional traits (e.g., plant height) can overwhelm interspecific differences in a restored grassland.*